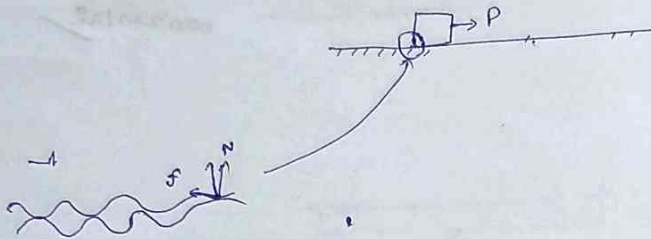
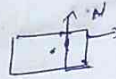
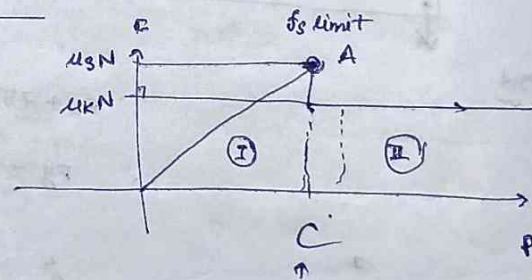
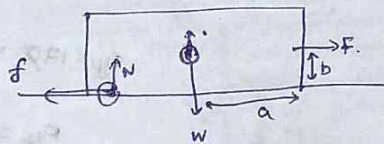
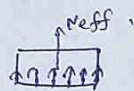


Friction

→ It is a force that resists the relative motion between two rigid bodies that are in contact with each other.



→ The resultant of the tangential component of the reaction force acting at the contact point (in microscopic level)



$$(I) \sum F_y = 0$$

$$N = W$$

$$(II) \sum F_x = 0$$

$$F = f$$

$$(III) F \cdot b - W \cdot a = N \cdot y$$

$$\Rightarrow \frac{F \cdot b - W \cdot a}{N} = y$$

Region 1 $\rightarrow$ Static condition
 Region 2 $\rightarrow$ Dynamic condition

$\mu_s \rightarrow$ coeff. of static friction

$\mu_k \rightarrow$ coeff. of kinetic friction

Note:- ① $P < \mu_s N \rightarrow$ Static condition

$P = \mu_s N \rightarrow$ verge of motion

$P > \mu_s N \rightarrow$ dynamic motion

$\rightarrow$ COULOMB'S LAW OF FRICTION

① along the surface
 surface, what will
 & change
 ② change the direction
 ③ & violations in law

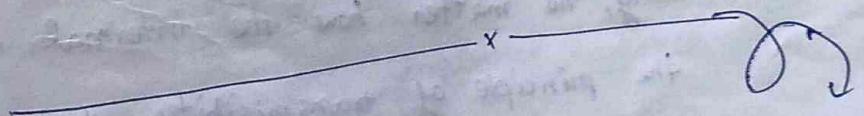
① The law has 3 assumptions:-

static friction

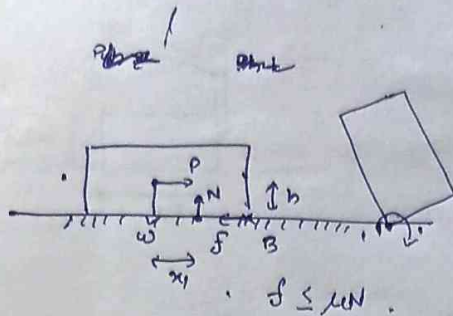
① The maximum force is directly proportional to the applied force

② The friction force b/w two bodies will not depend on the surface area, but depends on the nature of the surface.

③ The coefficient of kinetic friction or the dynamic friction will not depend on the magnitude of the velocity.



Sliding and Tipping



$$Ph + Nx_2 = W[x_1 + x_2]$$

$$\Rightarrow Ph + Wx_2 = Wx_1 + Wx_2$$

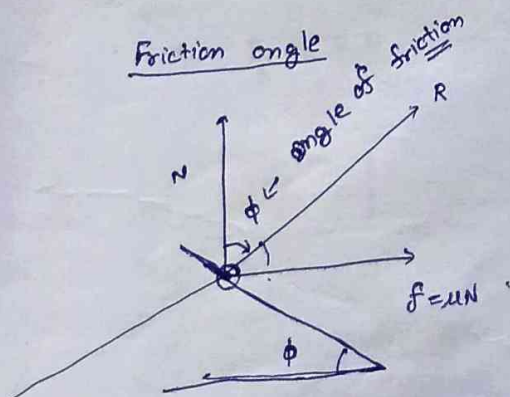
$$\Rightarrow Ph = Wx_1$$

$$x_1 \geq \frac{Ph}{W}$$

Condition to achieve tripping over sliding

- increase the magnitude of the applied force
- increase the value of h
- decrease the value of w .

Friction angle

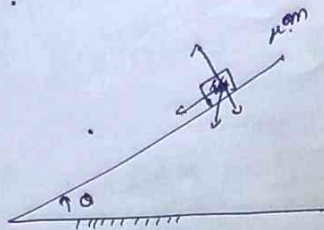


R = resultant of normal / friction force μN

$\phi \rightarrow$ friction angle
Angle b/w N force and frictional force + Normal force

$$\phi = \tan^{-1}(\mu)$$

Angle of repose

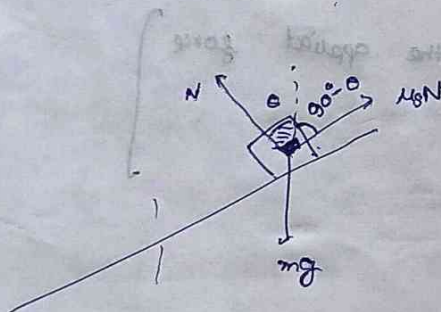


$\theta_R \rightarrow$ Magnitude of that maximum angle (inclination) at which the body will be in eq static equilibrium, without any slipping motion.

$$\mu mg \sin \theta = mg \sin \theta$$

$$\Rightarrow \mu = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1}(\mu)$$



LAMI'S rule

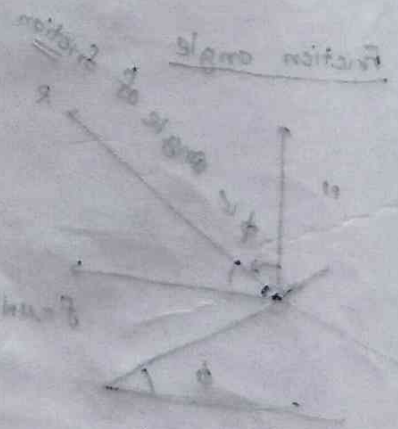
$$\frac{N \tan \theta}{\cos \theta} = \frac{\mu_s N}{\sin \theta}$$

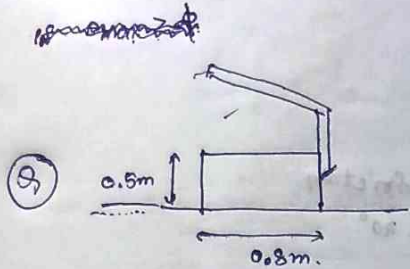
$$\Rightarrow \tan \theta = \mu_s$$

$$\Rightarrow \theta = \tan^{-1}(\mu_s)$$

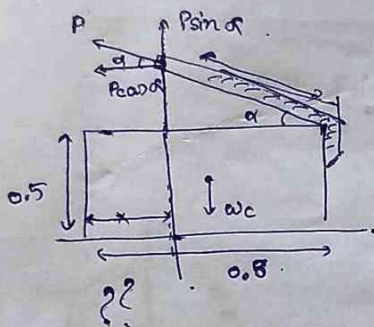
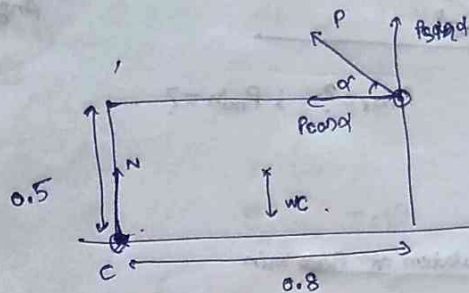
$\rightarrow$ if $\theta_{max} > \theta$, body is in ^{non} static equilibrium

$\rightarrow$ if $\theta_{max} \leq \theta$, static equilibrium exist





Find the minimum P and α to move



$$W_c \times 0.4 = P \sin \alpha \times 0.5$$

$$40 \times 0.4 = P \sin \alpha \times 0.5$$

$$P \sin \alpha = 32 \quad \text{--- (I)}$$

$$P \sin \alpha = \frac{4}{5} W_c$$

$$\Rightarrow P \cos \alpha \leq f \quad \text{--- (II)}$$

$$\Rightarrow P \sin \alpha + N = W_c$$

$$\Rightarrow W_c - P \sin \alpha = N \quad \text{--- (III)}$$

$$W_c - \frac{4}{5} W_c = N$$

$$\boxed{\frac{W_c}{5} = N}$$

$$P \cos \alpha = \mu_s \frac{W_c}{5}$$

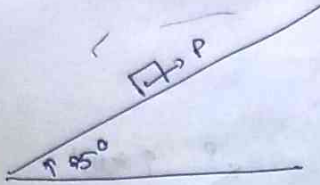
$$\frac{35}{140} = \frac{12}{2}$$

$$\tan \alpha = 4 \mu_s$$

$$\alpha = \tan^{-1}(1.4)$$

$$\boxed{\alpha = \tan^{-1}(4 \mu_s)}$$

9.

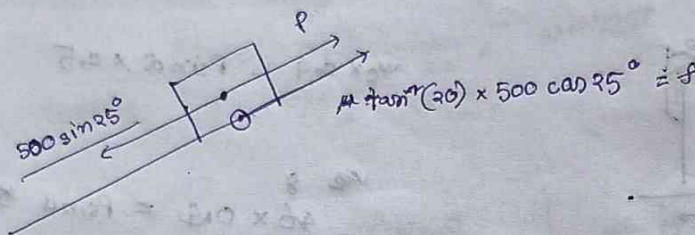


$w = 500N$

Angle of friction
 $= 20^\circ$

$P_{max} = ? ; P_{min} = ?$

Calculation of P_{min}



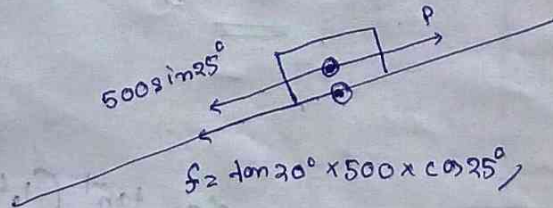
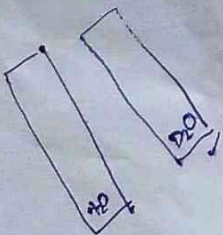
$P + \tan(20) \times 500 \cos 25^\circ = 500 \sin 25^\circ$

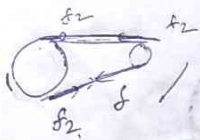
$P_{min} = 500 \sin 25^\circ - \tan(20) \times 500 \cos 25^\circ$

$= 500 \times 0.42 - 0.36 \times 500 \times 0.90$

$P_{min} = \boxed{48N} \approx \underline{46.37}$

Calculation of P_{max}





$$N = 2T$$

$$P_{min} = 500 \times \tan(20^\circ) \times \cos 25^\circ + 500 \sin 25^\circ$$

$$= 372N$$

$$= 376$$

$$\therefore (P_{min}, P_{max}) = [48, 372]$$

$$\frac{dT}{T} = \mu d\theta$$

$$\frac{dT}{T} = \mu d\theta$$

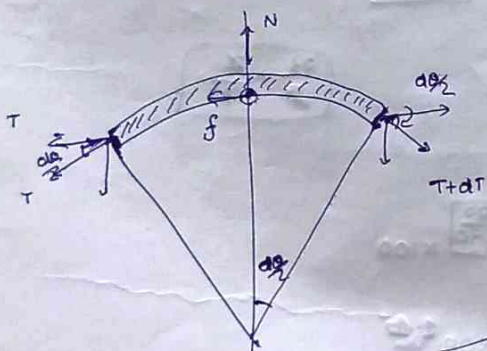
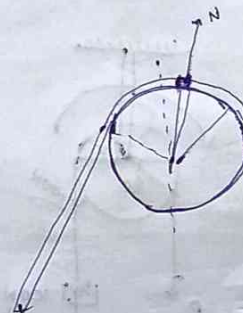
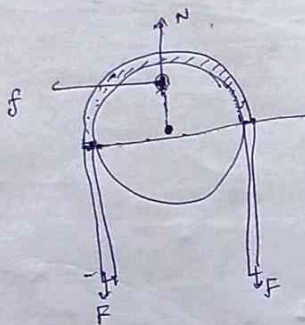
$$\int_{T_0}^{T_1} \frac{dT}{T} = \mu \int_0^{\theta} d\theta$$

$$\ln \left[\frac{T_1}{T_0} \right] = \mu \theta$$

$$T_1 = T_0 e^{\mu \theta}$$

$$\frac{T_2}{T_1} = e^{\mu \theta}$$

Belt Friction



$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$N = (T + dT + T) \sin \frac{d\theta}{2}$$

$$N = (2T + dT) \sin \frac{d\theta}{2}$$

$$N = T d\theta$$

$$\frac{1}{\mu} dT \cos \frac{d\theta}{2} = (2T + dT) \sin \frac{d\theta}{2}$$

$$dT \cos \frac{\theta}{2} = \mu N$$

$$dT \left[\frac{\cos \frac{d\theta}{2}}{\mu} - \sin \frac{d\theta}{2} \right] = 2T \sin \frac{d\theta}{2}$$

$$dT \cos \frac{\theta}{2} = \mu N$$

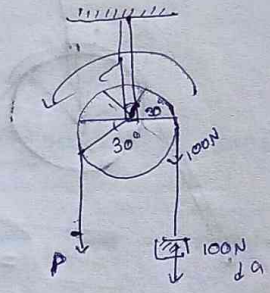
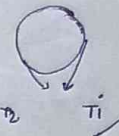
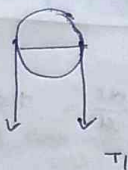
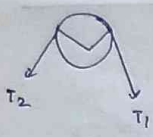
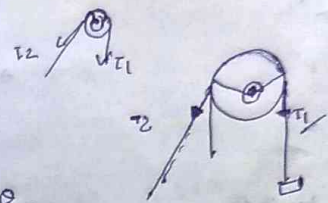
$$\frac{1}{\mu} dT \cos \frac{\theta}{2} = N$$

$$dT \left[\frac{1}{\mu} - \frac{d\theta}{2} \right] = 2T \cdot \frac{d\theta}{2}$$

$$\frac{T_2}{T_1} = e^{\mu\theta}$$

$$e^{\mu\theta} \geq 1$$

increase the value of μ or θ



$$\mu_s = 0.3$$

Solnⁿ:->

$$\frac{P}{100} = e^{0.3 \times \frac{7\pi}{6}}$$

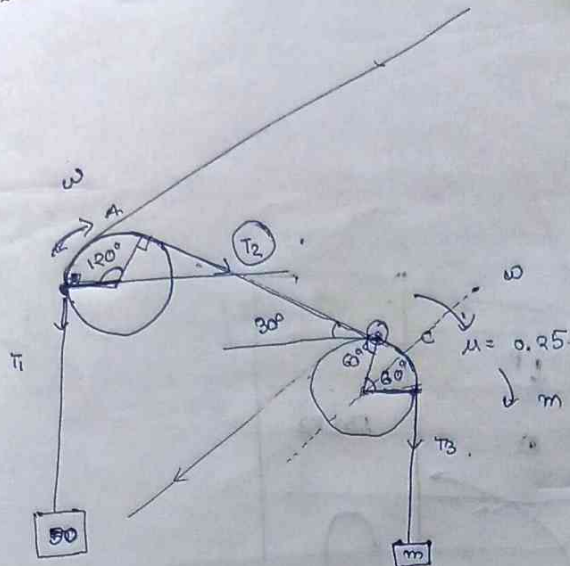
$$P = 100 e^{\frac{7\pi}{20}} = e^{\frac{39}{30}} \times 100 = 100 e^{1.3}$$

$$\frac{\pi + \pi}{6}$$

$$\frac{39}{30} \times \frac{7\pi}{20}$$

$$= 2.71 \times 100$$

$$= 271 = 300$$



Soln^o

$$\frac{T_2}{50} = e^{\frac{2\pi}{3} \times 0.25} \quad \text{--- (1)}$$

$$\frac{T_3}{T_2} = e^{\frac{\pi}{3} \times 0.25} \quad \text{--- (2)}$$

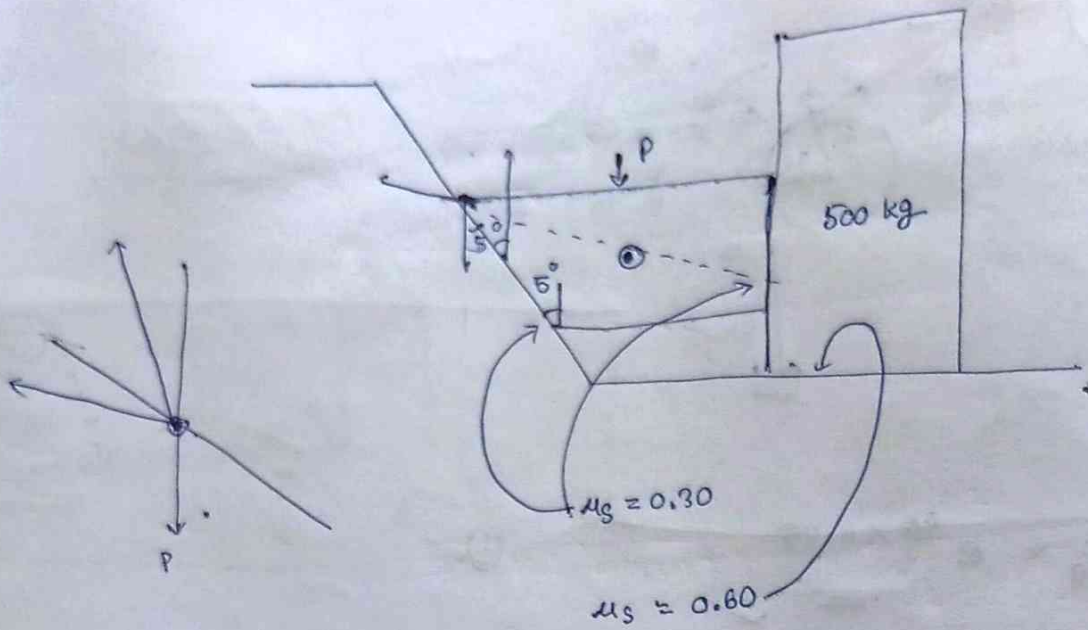
$$\frac{T_3}{50} = e^{\pi \times 0.25}$$

$$T_3 = 50 e^{\frac{3.14 \times 0.25}{1}}$$

$$m = 50 e^{\frac{3.14 \times 0.25}{1}}$$

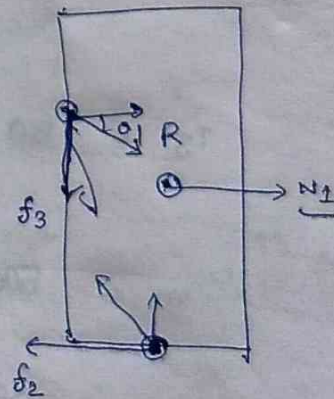
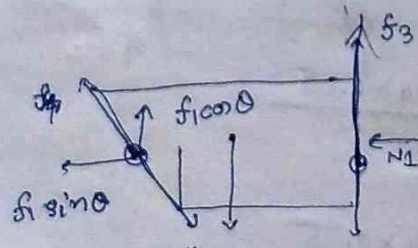
$$= 109.62 \text{ kg}$$

WEDGE - FRICTION



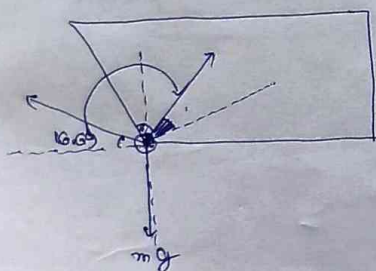
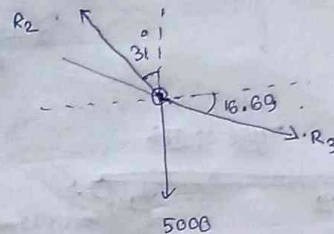
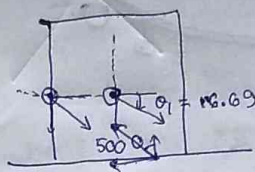
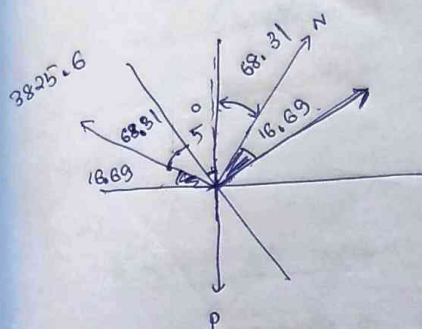
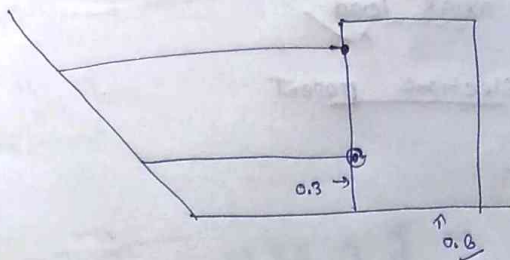
Q. calculate the $P_{\min}$ to move the block

Method 2: Use resultant method



$$0.60[f_3 + 5000] = (f_2)_{\max} \quad \text{--- (1)}$$

$$0.60[f_3 + 5000] = N_1$$



$$\frac{5000}{\sin(31^\circ)} = \frac{R_3}{\sin(31^\circ)}$$

$$3825.6 = R_3$$

$$\frac{P}{\sin[68.31 \times 2 + 5]} = \frac{3825.6}{\sin[68^\circ - 68.31 - 16.69]}$$

$$\frac{P}{\sin(141.62)} = \frac{3825.6}{\sin(95^\circ)}$$

$$P = 2,384.289 \text{ N}$$